# ME 361
# DYNAMICS OF MACHINERY

# UNIT 5
# BALANCING OF MACHINERY

*BSC MECHANICAL ENGINEERING, 3RD YEAR*

**INSTITUTE OF DISTANCE LEARNING/DEPARTMENT OF MECHANICAL ENGINEERING**
**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI**

# BALANCING

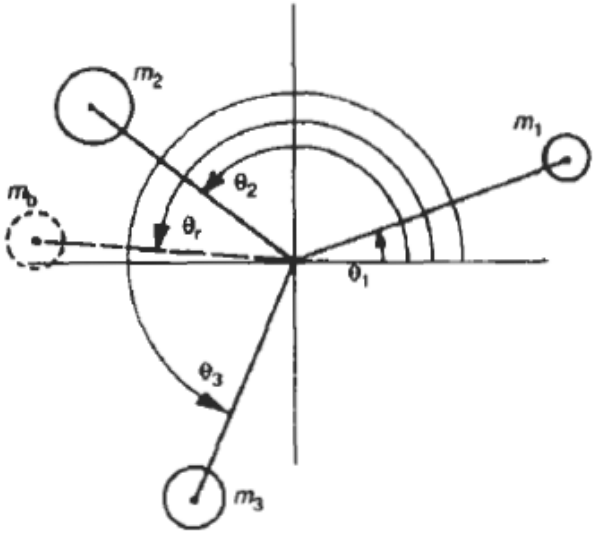
- ✓Balancing of rotating engines
  - Same plane (Static)
  - Different planes (Dynamic)
- ✓Balancing of reciprocating engines
  - Inline
  - Radial
    - V-Engines
    - W-Engines etc.

# Introduction

- Unbalance linkages produce shaking forces and torques, as was learnt in Unit 4. In this unit, we learn about design of linkages to eliminate shaking effects, which cause vibration in machine supports.
- The process of eliminating or minimizing shaking effects is called balancing of linkage. Balancing of linkages is more difficulty than balancing of rotors, and in many cases complete balance cannot be achieved by practical means.
- Many methods have been developed to balance linkages. Some of the methods achieve a complete balance of one dynamic factor, such as shaking force at the expense of shaking torque, or vice versa. Typically, this involves adding counterweights or/and redistributing the masses of links to change the locations of the centres of mass. This unit covers balancing of four-bar linkage to balancing of multi-cylinder engines.
- Complete balance of any mechanism may be obtained by connecting a mirror image mechanism to cancel the dynamic forces and moments. Certain configurations of multi-cylinder internal combustion (IC) engines and reciprocating compressors utilises this principle. As a result most multi-cylinder IC engines have even number of cylinders, such as four cylinders, six cylinders, eight cylinders, which are arranged such that shaking effect of cylinder cancels that of another.
- Increase in pin force and weight also result in an increase in shaking torque compared to the value on unbalance linkage. Balancing by addition of weight will increase mass and mass moment of inertia. As a result, the torque required to drive the linkage will be greater.

# Review-Balancing of rotating masses Same plane (Static)

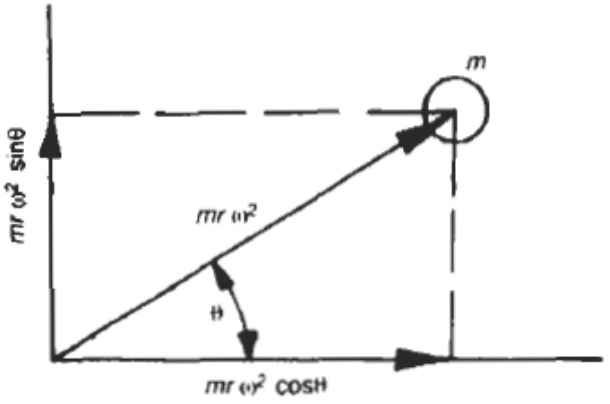
Balancing of rotating components is of extreme importance, especially in the *case* of high-speed machinery. Lack of balance may be due to masses in one plane or masses in different planes some distance apart. The method of balancing is given.




The forces are: $m_1\omega^2 r_1$, $m_2\omega^2 r_2$, etc. These are resolved into vertical and horizontal components:

$$F_V = m_1\omega^2 r_1 sin\theta_1 + m_2\omega^2 r_2 sin\theta_2 + \cdots$$

$$F_H = m_1\omega^2 r_1 cos\theta_1 + m_2\omega^2 r_2 cos\theta_2 + \cdots$$




Resultant force, $F = \sqrt{{F_V}^2 + {F_H}^2}$ at an angle to horizontal axis $\theta = \tan^{-1}\left(\frac{F_V}{F_H}\right)$

# How to Determine Angle

| $Fh$ | $Fv$ | $\theta$ |
|---|---|---|
| positive | positive | $\theta = \tan^{-1}\left(\frac{Fv}{Fh}\right)$ |
| negative | positive | $\theta = 180 - \tan^{-1}\left(\left\|\frac{Fv}{Fh}\right\|\right)$ |
| negative | negative | $\theta = 180 + \tan^{-1}\left(\left\|\frac{Fv}{Fh}\right\|\right)$ |
| positive | negative | $\theta = 360 - \tan^{-1}\left(\left\|\frac{Fv}{Fh}\right\|\right)$ |

# Review-Balancing of rotating masses Same plane (Static)

EXAMPLE

Four masses $m_1$, $m_2$, $m_3$ and $m_4$ are 200 kg, 300 kg, 240 kg and 260 kg respectively. The corresponding radii of rotation are 0.2 m, 0.15 m, 0.25 m and 0.3 m respectively and the angles between successive masses are 45°, 75° and 135°. Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2 m.

$$\sum F_V = 200 \times 0.2sin0 + 300 \times 0.15sin45 + 240 \times 0.25sin120 + 260 \times 0.3sin255$$

$$\sum F_V = \text{ 8.5 kg m}$$

$$\sum F_H = 200 \times 0.2cos0 + 300 \times 0.15cos45 + 240 \times 0.25cos120 + 260 \times 0.3cos255$$

$$\sum F_H = \text{ 21.6 kg m}$$

Unbalance force, $F = \sqrt{\sum F_V{}^2 + \sum F_H{}^2} = 23.2$ kg m $\qquad \theta = \tan^{-1}\left(\frac{8.5}{21.6}\right) = 21.48°$

# Review-Balancing of rotating masses Different planes (Dynamic)

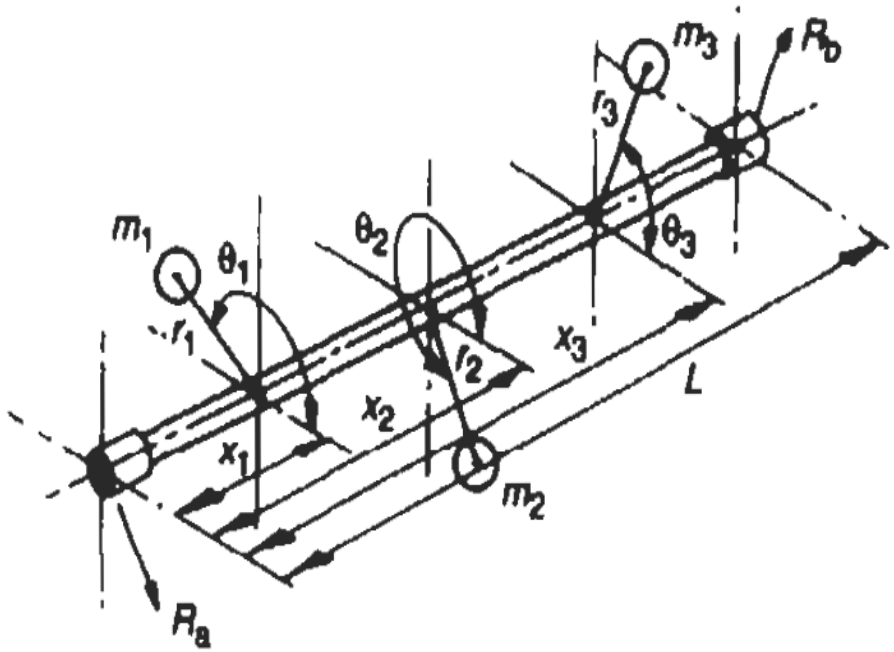
Method of *balancing* Complete dynamic balance *is* achieved by introducing forces equal and opposite to $R_a$, and $R_b$. In practice, balancing is carried out at planes a short distance from the bearings.

$$\sum F_V = \sum_{i=1}^{i=n} m_i \omega^2 r_i sin\theta_i = 0$$

$$\sum F_H = \sum_{i=1}^{i=n} m_i \omega^2 r_i cos\theta_i = 0$$

$$\sum M_a = \sum_{i=1}^{i=n} m_i \omega^2 r_i x_i sin\theta_i = 0$$

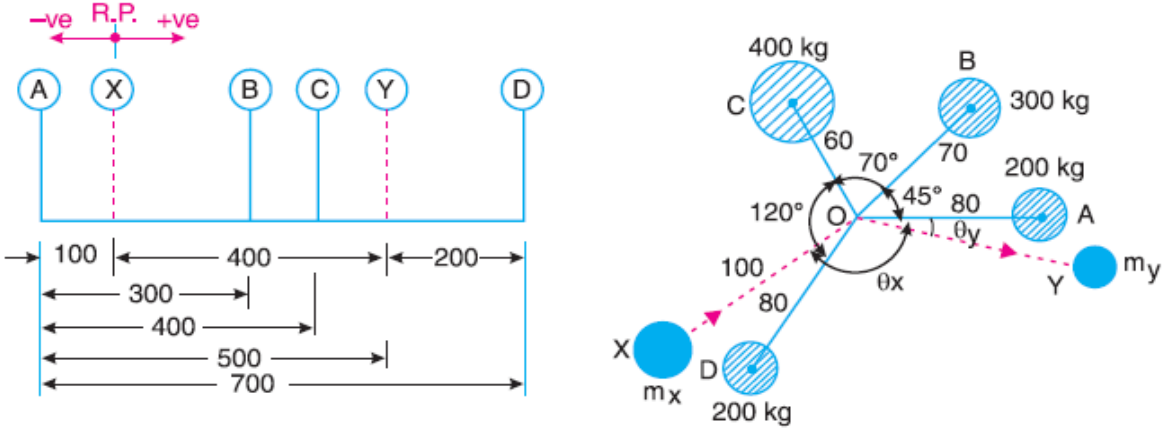
$$\sum M_b = \sum_{i=1}^{i=n} m_i \omega^2 r_i x_i cos\theta_i = 0$$



# Review-Balancing of rotating masses Different planes (Dynamic)

EXAMPLE

A shaft carries four masses A, B, C and D of magnitude 200 kg, 300 kg, 400 kg and 200 kg respectively and revolving at radii 80 mm, 70 mm, 60 mm and 80 mm in planes measured from A at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are A to B 45°, B to C 70° and C to D 120°. The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100 mm, between X and Y is 400 mm and between Y and D is 200 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions.




All dimensions in mm.

(*a*) Position of planes. (*b*) Angular position of masses.

# Review-Balancing of rotating masses Different planes (Dynamic)

| Plane | *kg* | *m* | *(m.r) kg-m* | *Plane x(l) m* | *(m.r.l) kg-m2* | $\theta^{\circ}$ | *(m.r) cos $\theta$* | *(m.r) sin $\theta$* | *(m.r.l) cos $\theta$* | *(m.r.l) sin $\theta$* |
|---|---|---|---|---|---|---|---|---|---|---|
| A | 200 | 0.08 | 16 | - 0.1 | - 1.6 | 0 | 16 | 0 | -1.6 | 0 |
| X(R.P.) | mx | 0.1 | 0.1 mx | 0 | 0 | $\theta_X$ | 0.1 mx cos$\theta_X$ | 0.1mxcos$\theta_X$ | 0 | 0 |
| B | 300 | 0.07 | 21 | 0.2 | 4.2 | 45 | 14.85 | 14.85 | 2.97 | 2.97 |
| C | 400 | 0.06 | 24 | 0.3 | 7.2 | 115 | -10.14 | 21.75 | -3.04 | 6.53 |
| Y | my | 0.1 | 0.1 my | 0.4 | 0.04 my | $\theta_Y$ | 0.1mycos$\theta_Y$ | 0.1mysin$\theta_Y$ | 0.04mycos$\theta_Y$ | 0.04mysin $\theta_Y$ |
| D | 200 | 0.08 | 16 | 0.6 | 9.6 | 235 | -9.18 | -13.11 | -5.51 | -7.86 |

$$\sum mrlcos\theta = 0.04\text{mycos}\theta_Y - 7.18 = 0;$$

$$\sum mrlsin\theta = 0.04\text{mysin}\theta_Y + 1.67 = 0;$$

$$\text{mycos}\theta_Y = 179.5;$$

$$\text{mysin}\theta_Y = -41.75;$$

$$my = \sqrt{179.5^2 + 41.75^2} = 184.29\ kg < -13.09°;$$

$$\sum mrcos\theta = 0.1\text{mxcos}\theta_X + 29.48 = 0;$$

$$\sum mrsin\theta = 0.1\text{mxsin}\theta_X - 19.32 = 0;$$

$$\text{mxcos}\theta_x = -294.8;$$

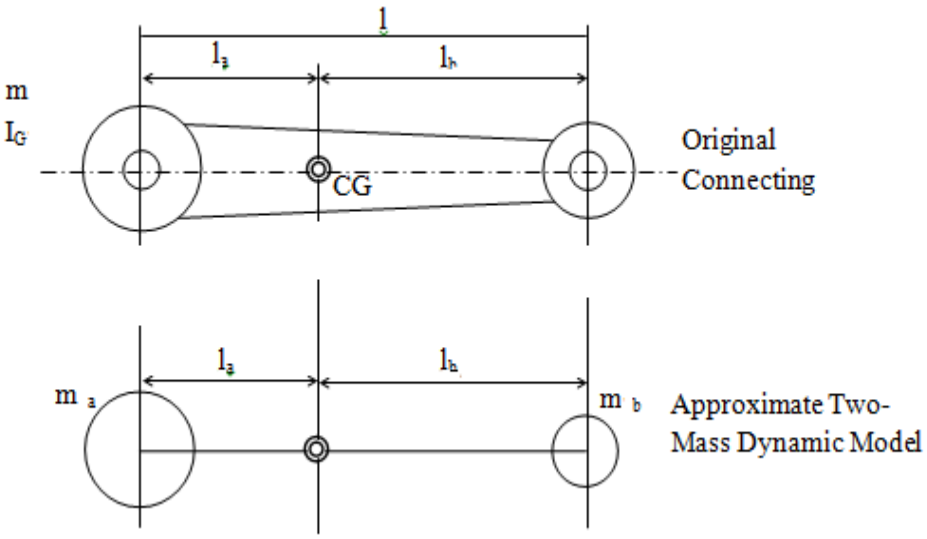
$$\text{mxsin}\theta_X = 193.2;$$

$$my = \sqrt{294.8^2 + 193.2^2} = 352.5\ kg < 146.76°;$$

# Lumped-Mass Dynamic Model of Connecting Rod

The model and actual connecting rods would by dynamically equivalent if the following three conditions are satisfied:

- The centres of mass must be at the same location, point $CG_2$.
- The total mass must be equal.
- The moment of inertia with respect to the centre of mass must be equal.




$$l = l_a + l_b$$

$$m = m_a + m_b$$

$$I_G = m_a l_a^{\,2} + m_b l_b^{\,2}$$

$$m_a l_a = m_b l_b$$

$$\therefore m_a = \frac{m l_b}{l_a + l_b}; m_b = \frac{m l_a}{l_a + l_b}$$

# Example 5-1

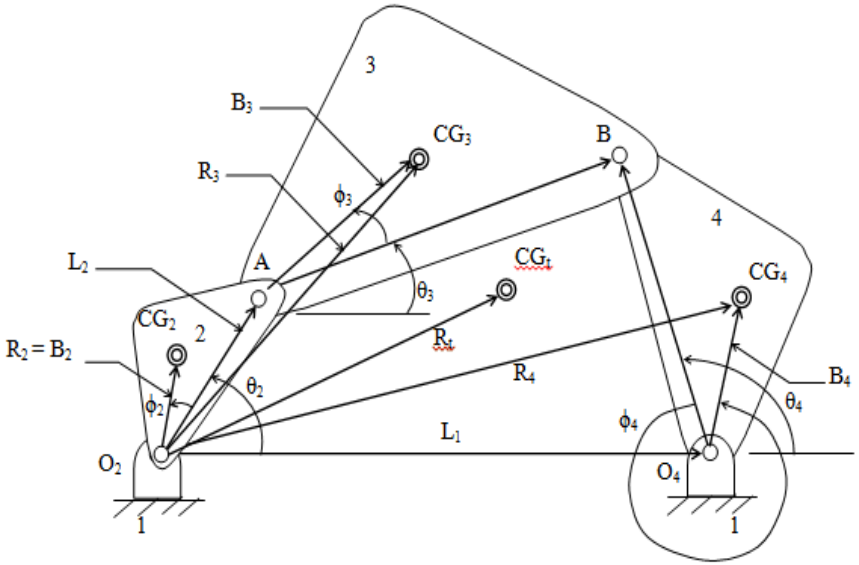
The distance between the connecting centres of a connecting rod of a single-cylinder two-stroke internal combustion engine of mass 8 kg is 240 mm. The mass moment of inertia of the connecting rod about an axis through the centre of mass is 6500 kg-mm$^2$, and its centre of mass is 90 mm from its large end centre. If one of the masses is to be located at the small end centre, determine the dynamical equivalent of the two lumped masses of the connecting rod.

$$m_a = m\frac{l_b}{l_a + l_b} = (8)\frac{(240-90)}{240} = 5\,\text{kg}$$

$$m_b = m\frac{l_a}{l_a + l_b} = (8)\frac{(90)}{240} = 3\,\text{kg}$$

# Balancing of Linkages

A rotating link of a linkage may be balanced using similar technique as the one for balancing rotating rotor. This requires making the centre of mass of the linkage stationary. This method works for any linkage having revolute (pin) and prismatic (slider) joints, provided that the prismatic joint is not connected to the ground. This means that this method is applicable to crank-slider machines such as internal combustion engines and reciprocating compressors.

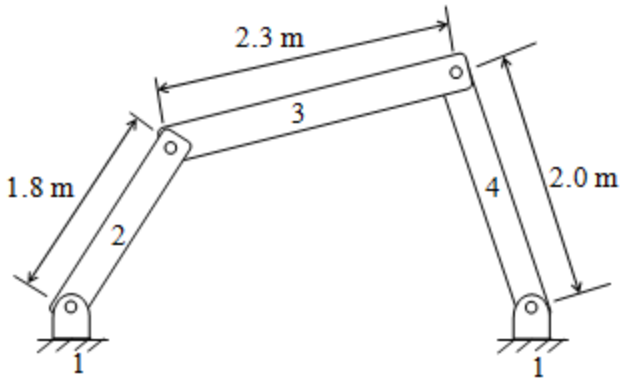



$$\left(m_2 B_2\right)_x = m_3\left(B_3 \frac{L_2}{L_3}\cos\phi_3 - L_2\right) \qquad \left(m_2 B_2\right)_y = m_3\left(B_3 \frac{L_2}{L_3}\sin\phi_3\right)$$

$$\left(m_4 B_4\right)_x = -m_3 B_3 \frac{L_4}{L_3}\cos\phi_3 \qquad \left(m_4 B_4\right)_y = -m_3 B_3 \frac{L_4}{L_3}\sin\phi_3$$

# Example 5-2

The masses of uniform cross section links 2, 3 and 4 shown are 0.5 kg, 1.0 kg and 1.2 kg, respectively. The angles between lines of centres and centres of mass of links 2, 3 and 4 are $7^o$, $10^o$ and $-3^o$, respectively. For complete force balance, determine the masses that must attached to link 2 at radius 0.6 m from the fixed joints $O_2$ and link 4 at radius 1.0 m from $O_4$, respectively. Also, determine the orientation of the centres of mass of the masses from their respective lines of centres of joints.




solution

$L_2 = 1.8\,\text{m}$ $\quad B_2 = \frac{L_2}{2} = 0.9\,\text{m}$ $\quad m_2 = 0.5\,\text{kg}$

$L_3 = 2.3\,\text{m}$ $\quad B_3 = \frac{L_3}{2} = 1.15\,\text{m}$ $\quad m_3 = 1.0\,\text{kg}$

$L_4 = 2.0\,\text{m}$ $\quad B_4 = \frac{L_4}{2} = 1.0\,\text{m}$ $\quad m_4 = 1.2\,\text{kg}$

$\phi_2 = 7^o$ $\quad \phi_3 = 10^o$ $\quad \phi_4 = 3^o$

# Example 5-2 cont'd…

$$(m_2 B_2)_x = m_3\left(B_3\frac{L_2}{L_3}\cos\phi_3 - L_2\right) = (1.0)\left[(1.15)\frac{(1.8)}{(2.3)}\cos(10) - 1.8\right]$$

$$(m_2 B_2)_x = -0.9137\ \text{kg - m}$$

$$(m_2 B_2)_y = m_3\left(B_3\frac{L_2}{L_3}\sin\phi_3\right) = (1.0)\left[(1.15)\frac{(1.8)}{(2.3)}\sin(10)\right]$$

$$(m_2 B_2)_y = 0.1563\ \text{kg - m}$$

$$(m_4 B_4)_x = -m_3 B_3\frac{L_4}{L_3}\cos\phi_3 = -(1.0)(1.15)\frac{(2.0)}{(2.3)}\cos(10)$$

$$(m_4 B_4)_x = -0.9848\ \text{kg - m}$$

$$(m_4 B_4)_y = -m_3 B_3\frac{L_4}{L_3}\sin\phi_3 = -(1.0)(1.15)\frac{(2.0)}{(2.3)}\sin(10)$$

$$(m_4 B_4)_y = -0.1736\ \text{kg - m}$$

Let $m_{2a}$ and $m_{4a}$ , and $m_{2c}$ and $m_{4c}$ be masses that should be attached to and original masses of links 2 and 4, respectively. Also, let $\phi_{2a}$ and $\phi_{4a}$ be orientation of the attached masses from the centre lines of links 2 and 4, respectively. Then

$$(m_{2a}B_{2a})\cos\phi_{2a} + (m_{2c}B_{2c})\cos\phi_{2c} = (m_2 B_2)_x$$

$$(m_{2a}B_{2a})\cos\phi_{2a} + [(0.5)(0.9)\cos(7)] = -0.9137$$

$$(m_{2a}B_{2a})\cos\phi_{2a} = -1.3603$$

$$(m_{2a}B_{2a})\sin\phi_{2a} + (m_{2c}B_{2c})\sin\phi_{2c} = (m_2 B_2)_y$$

$$(m_{2a}B_{2a})\sin\phi_{2a} + [(0.5)(0.9)\sin(7)] = 0.1565$$

$$(m_{2a}B_{2a})\sin\phi_{2a} = 0.10166$$

$$m_{2a}B_{2a} = \sqrt{(m_{2a}B_{2a}\cos\phi_{2a})^2 + (m_{2a}B_{2a}\sin\phi_{2a})^2} = \sqrt{(-1.3603)^2 + (0.10166)^2}$$

$$m_{2a}B_{2a} = m_{2a}(0.6) = 1.3641\ \text{kg - m} \qquad m_{2a} = 2.2735\ \text{kg}$$

$(m_{2a}B_{2a})\cos\phi_{2a}$ $(m_{2a}B_{2a})\sin\phi_{2a}$

$$90 < \phi_{2a} < 180$$

$$\phi_{2a} = 180 - \tan^{-1}\left(\left|\frac{(m_{2a}B_{2a})\sin\phi_{2a}}{(m_{2a}B_{2a})\cos\phi_{2a}}\right|\right) = 180 - \tan^{-1}\left(\left|\frac{0.10166}{-1.3603}\right|\right)$$

$$\phi_{2a} = 175.7^o$$

# Example 5-2 cont'd…

Similarly for link 4, we have

$$(m_{4a}B_{4a})\cos\phi_{4a} + (m_{4c}B_{4c})\cos\phi_{4c} = (m_4B_4)_x$$

$$(m_{4a}B_{4a})\cos\phi_{4a} + [(1.2)(1.0)\cos(-3)] = -0.9848$$

$$(m_{4a}B_{4a})\cos\phi_{4a} = -2.1832$$

$$(m_{4a}B_{4a})\sin\phi_{4a} + (m_{4c}B_{4c})\sin\phi_{4c} = (m_4B_4)_y$$

$$(m_{4a}B_{4a})\sin\phi_{4a} + [(1.2)(1.0)\sin(-3)] = -0.1736$$

$$(m_{4a}B_{4a})\sin\phi_{2a} = -0.1108$$

$$m_{4a}B_{4a} = \sqrt{(m_{4a}B_{4a}\cos\phi_{4a})^2 + (m_{4a}B_{4a}\sin\phi_{4a})^2} = \sqrt{(-2.1832)^2 + (0.1108)^2}$$

$$m_{4a}B_{4a} = m_{4a}(1.0) = 2.186\ \text{kg - m}$$

$$m_{4a} = 2.186\ \text{kg}$$

Since both

$(m_{4a}B_{4a})\cos\phi_{4a}$ and $(m_{4a}B_{4a})\sin\phi_{4a}$

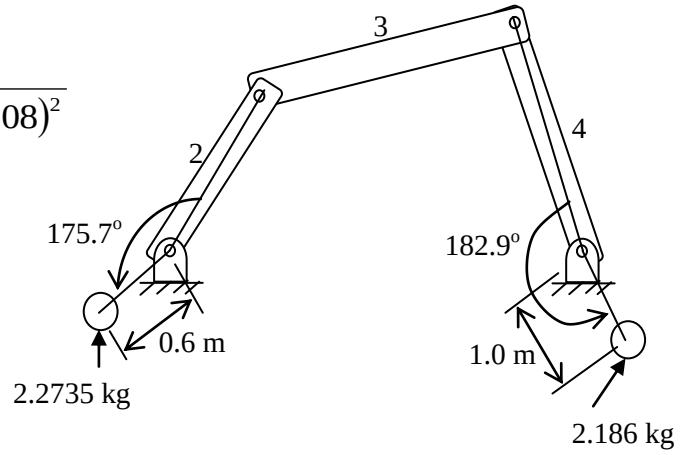
are negative, $\phi_{4a}$ lies within

$$180 < \phi_{2a} < 270$$

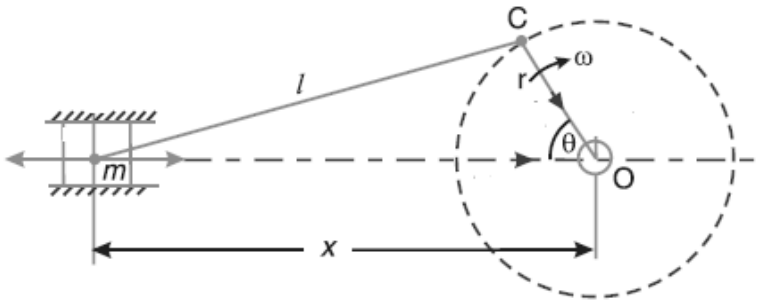
$$\phi_{2a} = 180 + \tan^{-1}\left(\left|\frac{(m_{4a}B_{4a})\sin\phi_{4a}}{(m_{4a}B_{4a})\cos\phi_{4a}}\right|\right) = 180 + \tan^{-1}\left(\left|\frac{-0.1108}{-2.1832}\right|\right)$$

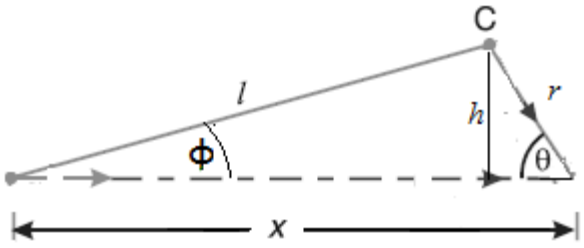
$$\phi_{2a} = 182.9^o$$ to the line centres of link 4

The location of masses is as shown below



# Balancing of Inline (reciprocating) Engines




$$h = r\sin\theta = l\sin\phi$$

$$x = r\cos\theta + l\cos\phi$$

$$\cos^2\phi + \sin^2\phi = 1 \Rightarrow \cos\phi = \sqrt{1-\left(\frac{r}{l}\sin\theta\right)^2}$$

$$\therefore x = r\cos\theta + l\sqrt{1-\left(\frac{r}{l}\sin\theta\right)^2}$$

$$recall; (1+x)^n = 1 + xn + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots + x^n$$

$$x = r\cos\theta + l\left(1 - \frac{1}{2}\left(\frac{r}{l}\sin\theta\right)^2 + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\left(\frac{r}{l}\sin\theta\right)^4 + \ldots - \frac{r}{l}\sin\theta\right)$$

$$\dot{x} = -r\omega\sin\theta - \frac{r^2}{2l}\omega\sin 2\theta; \textit{taking the first two items of the series}$$

$$\ddot{x} = -r\omega^2\left(\cos\theta + \frac{1}{n}\cos 2\theta\right)$$

$\therefore$ the unbalance (shaking) force on one cylinder is

$$F = m\ddot{x} = m\omega^2 r(\cos\theta + \frac{1}{n}\cos 2\theta); \text{ where } n = \frac{l}{r}$$

# Balancing of inline (reciprocating) engines cont'd…

For more than one cylinder

$$F = \omega^2 \sum_{i=1}^{i=z} m_i r_i \left\{ \cos(\theta + \phi_i) + \frac{1}{n} \cos(2\theta + 2\phi_i) \right\};$$

$$\sum_{i=1}^{i=z} m_i r_i \sin\phi_i = 0; \sum_{i=1}^{i=z} m_i r_i \cos\phi_i = 0, \text{conditions for primary force balance}$$

$$\sum_{i=1}^{i=z} m_i r_i \sin 2\phi_i = 0; \sum_{i=1}^{i=z} m_i r_i \cos 2\phi_i = 0, \text{conditions for secondary force balance}$$

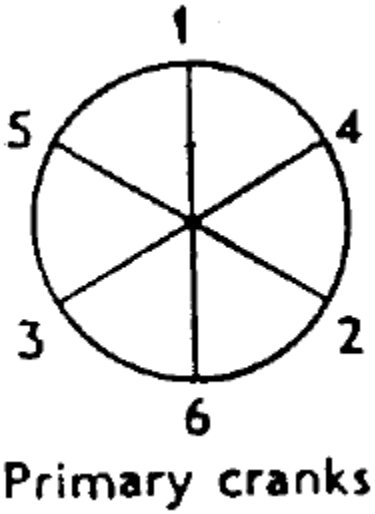
$$\sum_{i=1}^{i=z} m_i r_i x_i \sin\phi_i = 0; \sum_{i=1}^{i=z} m_i r_i x_i \cos\phi_i = 0, \text{conditions for primary couple balance}$$

$$\sum_{i=1}^{i=z} m_i r_i x_i \sin 2\phi_i = 0; \sum_{i=1}^{i=z} m_i r_i x_i \cos 2\phi_i = 0, \text{conditions for secondary couple balance}$$

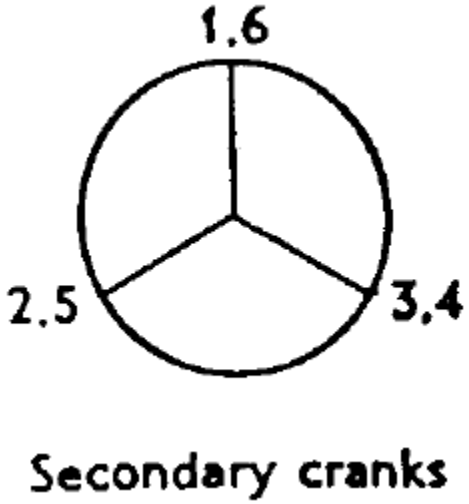
# Example 5-3

The six cylinders of a single-acting, two-stroke cycle Diesel engine are pitched 1 m apart and the cranks are spaced at 60° intervals. The crank length is 300 mm and the ratio of connecting rod to crank is 4.5. The reciprocating mass per line is 1.35 Mg and the rotating mass is 1 Mg. The speed is 200 rev/min. Show with regard to primary and secondary; balance, that the firing order 1-5-3-6-2-4 gives unbalance in primary moment only, and the order 1-4-5-2-3-6 gives secondary moment unbalance only. Compare the maximum values of these moments, evaluating them with respect to the central plane of the engine.

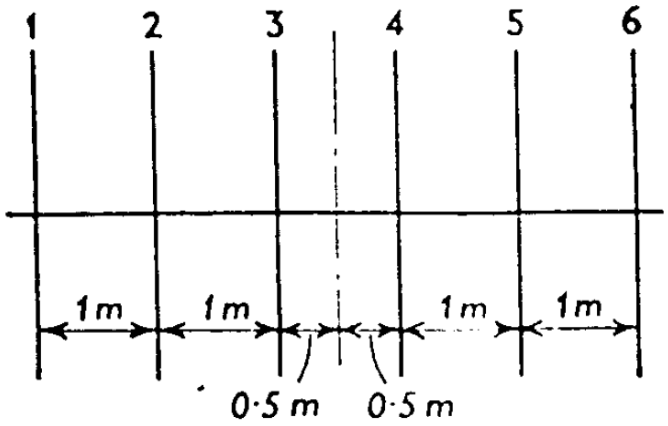
Firing Order 1-5-3-6-2-4.



Primary cranks



Secondary cranks



# Example 5-3 conti;

$$\sum sin\emptyset_i = sin0 + sin60 + sin120 + sin180 + sin240 + sin300 = 0$$

$$\sum cos\emptyset_i = cos0 + cos60 + cos120 + cos180 + cos240 + cos300 = 0$$

$$\sum sin2\emptyset_i = sin0 + sin120 + sin240 + sin360 + sin480 + sin600 = 0$$

$$\sum cos2\emptyset_i = cos0 + cos120 + cos240 + cos360 + cos480 + cos600 = 0$$

Therefore the primary and secondary forces are balanced

$$\sum x_i sin\emptyset_i = -2.5sin0 + 1.5sin60 - 0.5sin120 + 2.5sin180 - 1.5sin240 + 0.5sin300 = \sqrt{3}$$

$$\sum x_i cos\emptyset_i = -2.5cos0 + 1.5cos60 - 0.5cos120 + 2.5cos180 - 1.5cos240 + 0.5cos300 = -3$$

Primary couple is not balanced

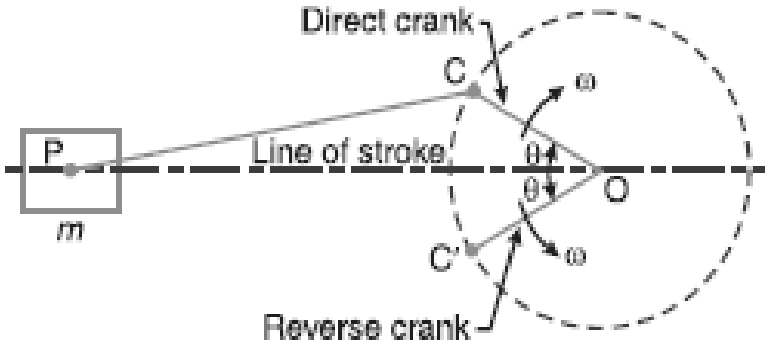
$$\sum x_i sin2\emptyset_i = -2.5sin0 + 1.5sin120 - 0.5sin240 + 2.5sin360 - 1.5sin480 + 0.5sin600 = 0$$

$$\sum x_i cos2\emptyset_i = -2.5cos0 + 1.5cos120 - 0.5cos240 + 2.5cos360 - 1.5cos480 + 0.5cos600 = 0$$

Secondary couples are balanced

Unbalance couple, $F = -(m + M)\omega^2 r\left(\sqrt{3}sin\theta + 3cos\theta\right) = 1.07\cos(\theta - 30°)\,MNm$

# Direct and Reverse Cranks




- Consider the diagram above, if the masses at C should be made equivalent to m/2
- Resolving the horizontal components of this force system gives

$$F_H = \frac{m}{2}\omega^2 r\cos\theta \times 2 = m\omega^2 r\cos\theta$$

# Direct and Reverse Cranks cont'd…

- **The vertical component**

$$F_V = \frac{m}{2}\omega^2 r\sin\theta - \frac{m}{2}\omega^2 r\sin\theta = 0$$

- **The secondary vertical force**

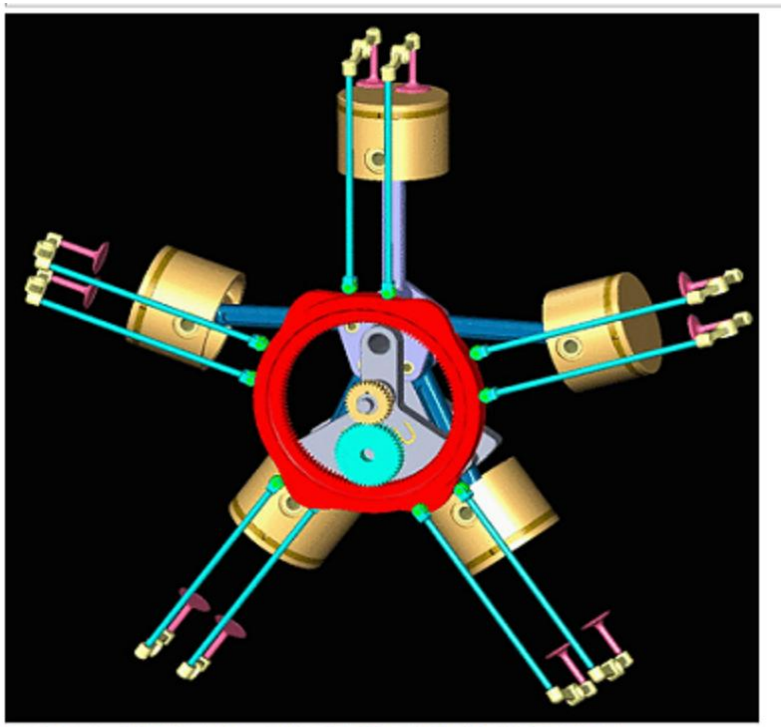
$$F_H = \frac{1}{n}m\omega^2 r\cos 2\theta$$

- **Similarly, the vertical component of the secondary force is zero**

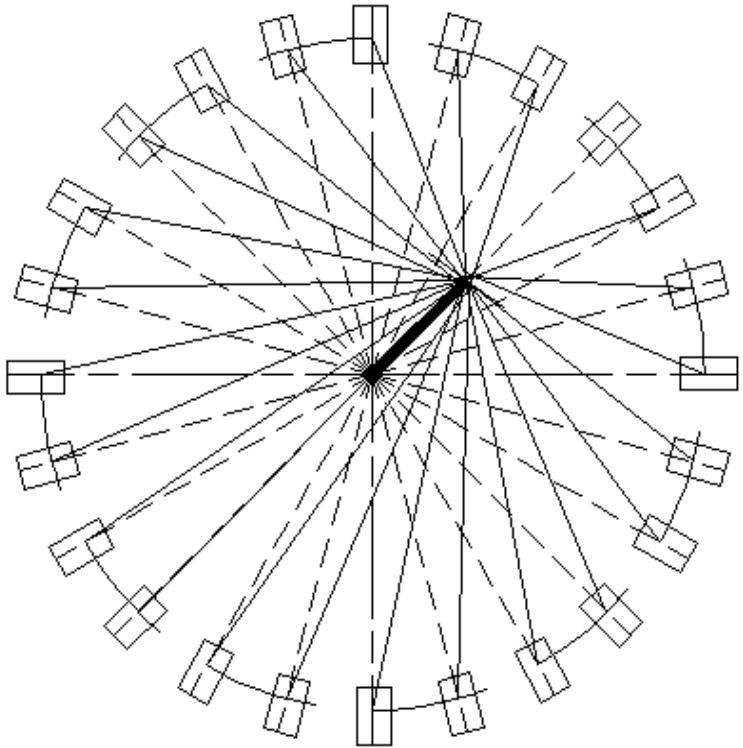
# RADIAL ENGINES

The radial engine is a reciprocating type internal combustion engine configuration in which the cylinders point outward from a central crankshaft. The radial configuration was commonly used in aircraft engines before turbine engines became predominant.

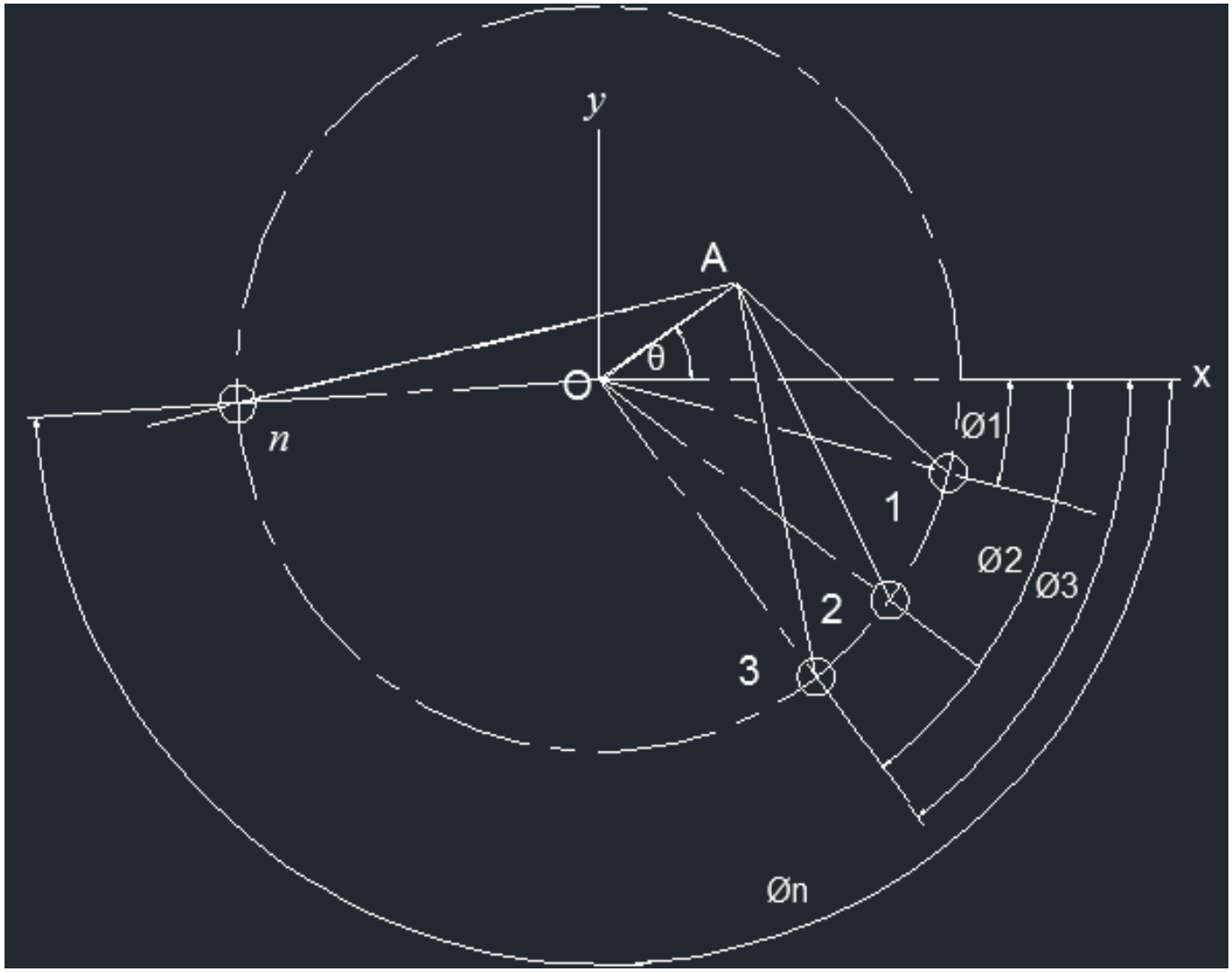
## R-5



## R-24



# Balancing of Radial Engines



# Balancing of radial engines cont'd…

*Total* Horizontal Primary Force

$$F_{px} = \omega^2 \sum_{i=1}^{i=z} m_i r_i \{\cos(\theta + \phi_i) \cos \phi_i\};$$

*Total* Vertical Primary Force

$$F_{py} = \omega^2 \sum_{i=1}^{i=z} m_i r_i \{\cos(\theta + \phi_i) \sin \phi_i\};$$

Resultant Primary Force

$$R_p = \sqrt{F_x{}^2 + F_y{}^2}$$

# Balancing of radial engines cont'd…

*Total* Horizontal Secondary Force

$$F_{sx} = \omega^2 \sum_{i=1}^{i=z} m_i r_i \left\{ \frac{1}{n} \cos(2\theta + 2\phi_i) \cos\phi_i \right\};$$

*Total* Vertical Secondary Force

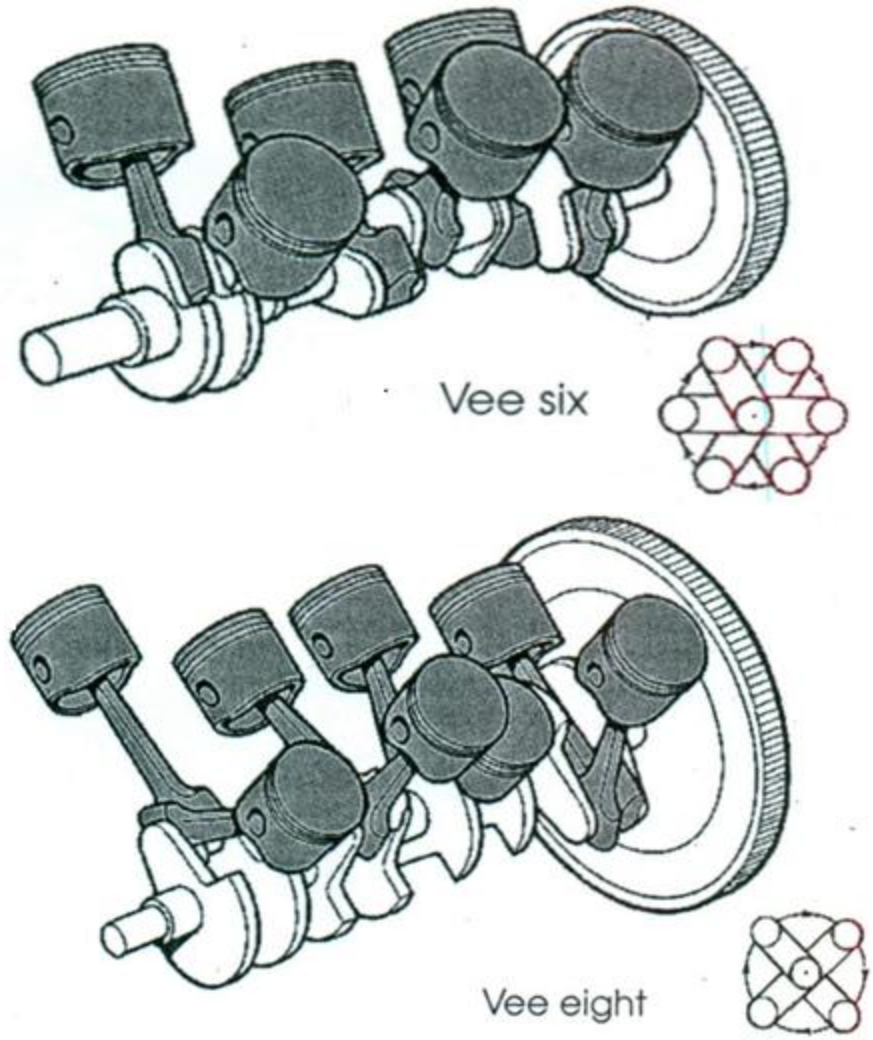
$$F_{sy} = \omega^2 \sum_{i=1}^{i=z} m_i r_i \left\{ \frac{1}{n} \cos(2\theta + 2\phi_i) \sin\phi_i \right\};$$

Resultant Secondary Force

$$R_s = \sqrt{F_x^{\,2} + F_y^{\,2}}$$

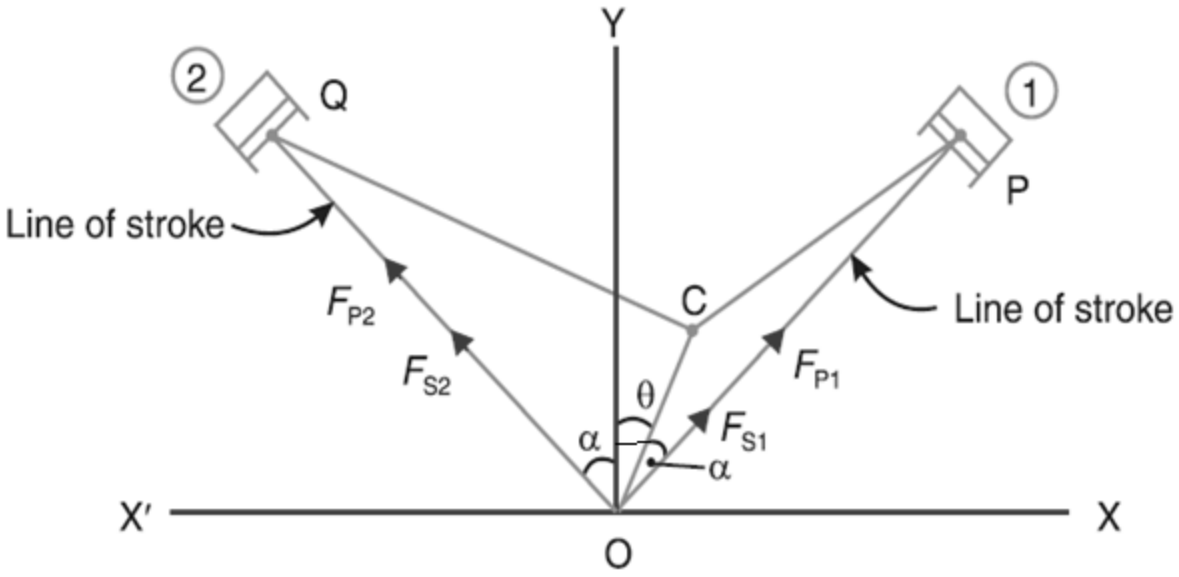
# V-ENGINES

The V-engine configuration is made up of two inline configurations with one on each bank, as shown. A V-6 engine is a six-cylinder engine which essentially consists of two three-cylinder inline engines on a common crankshaft. Similarly, a V-8 engine consists of two-cylinder inline engines on a common crankshaft. In V configurations, the bank angle introduces additional phase shift of the forces.



# Balancing of V-Engines

Consider a two-cylinder engine with one cylinder in each bank and with both sharing a common crank throw, and each cylinder inclined at an angle α to the vertical. By letting the y-axis be the central axis of the banks, the x-axis being the horizontal, with angle θ measured from the piston axis, the shaking force along x and y can be calculated.



# Balancing of V-Engines

*Total* primary vertical force

$$F_V = m\omega^2 r\cos(\alpha-\theta)\cos\alpha + m\omega^2 r\cos(\alpha+\theta)\cos\alpha$$

$$F_V = 2m\omega^2 r\cos^2\alpha\cos\theta$$

*Total* primary horizontal force

$$F_H = m\omega^2 r\cos(\alpha-\theta)\sin\alpha - m\omega^2 r\cos(\alpha+\theta)\sin\alpha$$

$$F_H = 2m\omega^2 r\sin^2\alpha\sin\theta$$

*R*esultant primary force

$$F_{PR} = \sqrt{F_H{}^2 + F_V{}^2}$$

$$F_{PR} = 2m\omega^2 r\sqrt{(\sin^2\alpha\sin\theta)^2 + (\cos^2\alpha\cos\theta)^2}$$

# Balancing of V-Engines

Similarly the secondary unbalance forces are

$$F_H = \frac{2m\omega^2 r}{n}(\sin\alpha \sin 2\alpha \sin 2\theta)$$

$$F_V = \frac{2m\omega^2 r}{n}(\cos\alpha \cos 2\alpha \cos 2\theta)$$

$$F_{SR} = \frac{2m\omega^2 r}{n}\sqrt{(\sin\alpha \sin 2\alpha \sin 2\theta)^2 + (\cos\alpha \cos 2\alpha \cos 2\theta)^2}$$

# Balancing of V-Engines cont'd…

For V - Engines with more than two cylinders

*Total* primary force on the left bank

$$F_{PL} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \left\{ \cos\beta_i \cos(\theta + \alpha) + \sin\beta_i \sin(\theta + \alpha) \right\}$$

*Total* primary force on the right bank

$$F_{PR} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \left\{ \cos\beta_i \cos(\alpha - \theta) + \sin\beta_i \sin(\alpha - \theta) \right\}$$

*Total* secondary force on the left bank

$$F_{SL} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \left\{ \cos 2\beta_i \cos(2\theta + 2\alpha) + \sin 2\beta_i \sin(2\theta + 2\alpha) \right\}$$

*Total* secondary force on the right bank

$$F_{SR} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \left\{ \cos 2\beta_i \cos(2\alpha - 2\theta) + \sin 2\beta_i \sin(2\alpha - 2\theta) \right\}$$

*where* $\beta_i$ is the 'i' th angle a crank makes with a reference.

# Example 5-4

The reciprocating mass per cylinder in a 60° V-twin engine is 1.5 kg. The stroke and connecting rod length are 100 mm and 250 mm respectively. If the engine runs at 2500 rpm., determine the maximum and minimum values of the primary and secondary forces. Also find out the crank position corresponding to these values.

# Example 5-4: analytical method

**Given 2α = 60° or α = 30°, m = 1.5 kg ; Stroke = 100 mm or r =** = 50 mm ; l = 250 mm ; N = 250 r.p.m. or $\omega = 2\pi \times 2500 / 60 = 261.8$ rad/s

**Maximum and minimum values of primary forces**

We know that the resultant primary force,

$$F_{PR} = 2m\omega^2 r\sqrt{(\sin^2\alpha\sin\theta)^2 + (\cos^2\alpha\cos\theta)^2}$$

$$F_{PR} = 2m\omega^2 r\sqrt{(\frac{3}{4}\cos\theta)^2 + (\frac{1}{4}\sin\theta)^2}$$

The primary force is maximum, when θ = 0°. Therefore, **maximum primary force is**

$$F_{PR} = 2\times 1.5\times (261.8)^2 \times 0.05\sqrt{(\frac{3}{4})^2} = 7.711\,kN$$

# Example 5-4 cont'd…

The primary force is minimum, when θ = 90°. Therefore ***the minimum primary force is***

$$F_{PR} = 2 \times 1.5 \times (261.8)^2 \times 0.05 \sqrt{(\frac{1}{4})^2} = 2.570\, kN$$

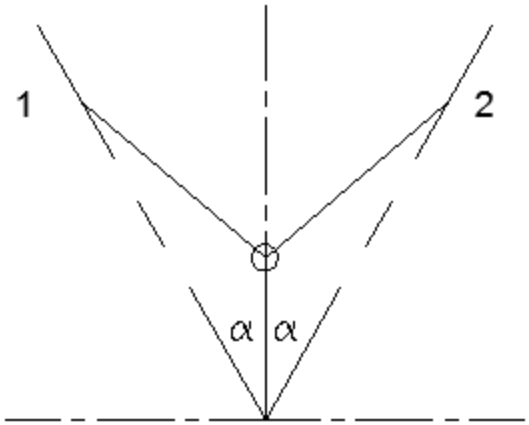
***Maximum and minimum values of secondary forces***

$$F_{SR} = \frac{2m\omega^2 r}{n} \sqrt{(\sin\alpha \sin 2\alpha \sin 2\theta)^2 + (\cos\alpha \cos 2\alpha \cos 2\theta)^2}$$

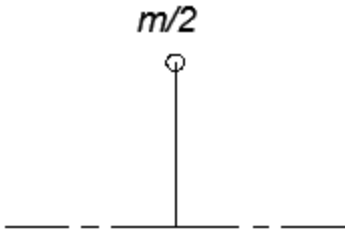
$$F_{SR} = \frac{2m\omega^2 r}{n} \sqrt{(\frac{1}{2} \times \frac{\sqrt{3}}{2} \sin 2\theta)^2 + (\frac{1}{2} \times \frac{\sqrt{3}}{2} \cos 2\theta)^2}$$

$$F_{SR} = \frac{\sqrt{3} \times 1.5 \times 261.8^2 \times 0.05}{2 \times (0.25/0.05)} = 890.3\ \text{N}$$

# Example 5-4: Graphical method

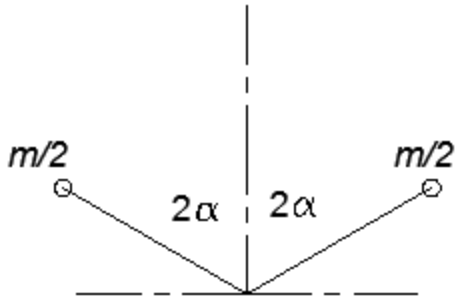



Arrangement of the Cylinders and Crank



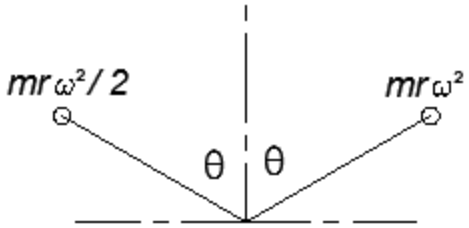

Pr*imary* Direct Force

$$F_{PD} = 2\frac{m}{2}\omega^2 r = m\omega^2 r$$




Pr*imary* Reverse Force

$$F_{PR} = 2\frac{m}{2}\omega^2 r\cos 60 = \frac{m}{2}\omega^2 r$$




*T*otal Horizontal Primary Force

$$F_{PH} = \frac{m}{2}\omega^2 r\sin\theta$$

*T*otal Vertical Primary Force

$$F_{PV} = \frac{3m}{2}\omega^2 r\cos\theta$$

# Example 5-4: Graphical method ( Solution) cont'd…

Resultant Primary Force

$$F_{PR} = \sqrt{F_{PV}{}^2 + F_{PH}{}^2} = 2m\omega^2 r\sqrt{(\frac{3}{4}\cos\theta)^2 + (\frac{1}{4}\sin\theta)^2}$$

*as* before.

Similarly, the Resultant Secondary Force

$$F_{SR} = \frac{2m\omega^2 r}{n}\sqrt{(\frac{1}{2}\times\frac{\sqrt{3}}{2}\sin 2\theta)^2 + (\frac{1}{2}\times\frac{\sqrt{3}}{2}\cos 2\theta)^2}$$

# Example 5-5

Investigate the out-of-balance forces of an eight-cylinder V-engine consisting of two banks of cylinders, each having four cylinders in line and both working upon one four-throw crankshaft. The centre-lines of the two banks are inclined at angles (Ø/2) on each side of the vertical plane. The relative positions of the four cranks are 0°, 180°, 180° and 0° and two connecting rods work on each crank. Find the maximum values of the horizontal and vertical forces acting on the engine, in terms of the angle Ø, the angular velocity ω of the crankshaft, the crank radius r, the connecting rod length l, and the reciprocating mass m per cylinder. State the nature and amount of the total force, (a) when Ø = 90°, (b) when Ø= 60° (c) Ø = 45°

# Example 5-5:analytical method

$$\sum sin\beta_i = sin0 + sin180 + sin180 + sin0 = 0$$

$$\sum cos\beta_i = cos0 + cos180 + cos180 + cos0 = 0$$

Primary forces are balanced

$$\sum sin2\beta_i = sin0 + sin360 + sin360 + sin0 = 0$$

$$\sum cos2\beta_i = cos0 + cos360 + cos360 + cos0 = 4$$

Secondary forces are not balanced

*Total* secondary force on the left bank

$$F_{SL} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \{\cos 2\beta_i \cos(2\theta + 2\alpha) + \sin 2\beta_i \sin(2\theta + 2\alpha)\}$$

*Total* secondary force on the right bank

$$F_{SR} = \omega^2 \sum_{i=1}^{i=n} m_i r_i \{\cos 2\beta_i \cos(2\alpha - 2\theta) + \sin 2\beta_i \sin(2\alpha - 2\theta)\}$$

# Example 5-5:analytical method

$$F_{SL} = 4mr\omega^2 \cos(2\theta + 2\alpha)$$

$$F_{SR} = 4mr\omega^2 \cos(2\alpha - 2\theta)$$

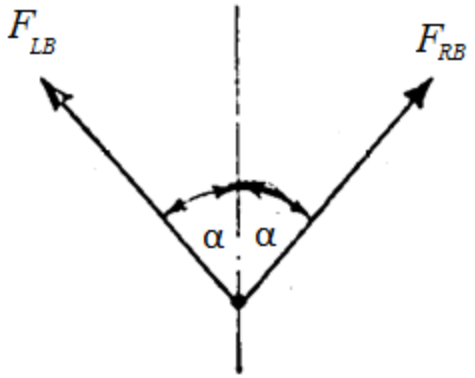
Total Vertical Force

$$\mathrm{F_V} = 8mr\omega^2 \cos 2\alpha \cos 2\theta \cos\alpha$$

Total Horizontal Force

$$\mathrm{F_H} = 8mr\omega^2 \sin 2\alpha \sin 2\theta \sin\alpha$$

Resultant Force




$$\mathrm{F_R} = \frac{8mr\omega^2}{n}\sqrt{(\cos 2\alpha \cos 2\theta \cos\alpha)^2 + (\sin 2\alpha \sin 2\theta \sin\alpha)^2}$$

$$when\ \phi = 90^\circ \qquad \mathrm{F_R} = \frac{4\sqrt{2}\, m\omega^2 r^2}{l} \sin 2\theta, \text{ sinusoidal}$$

$$when\ \phi = 60^\circ \qquad \mathrm{F_R} = \frac{2\sqrt{3}\, m\omega^2 r^2}{l}, \text{ constant}$$

$$when\ \phi = 45^\circ \qquad \mathrm{F_R} = ??$$

# 3,4,5 ASSIGNMENT 1

A shaft carries five masses *A*, *B*, C, D and *E* which revolve at the same radius in planes which are equidistant from one another. The magnitude of the masses in planes *A*, C and D are 50 kg, 40 kg and 80 kg respectively. The angle between *A* and C is 90° and that between C and D is 135°. Determine the magnitude of the masses in planes *B* and *E* and their positions to put the shaft in complete rotating balance.

# ASSIGNMENT 2

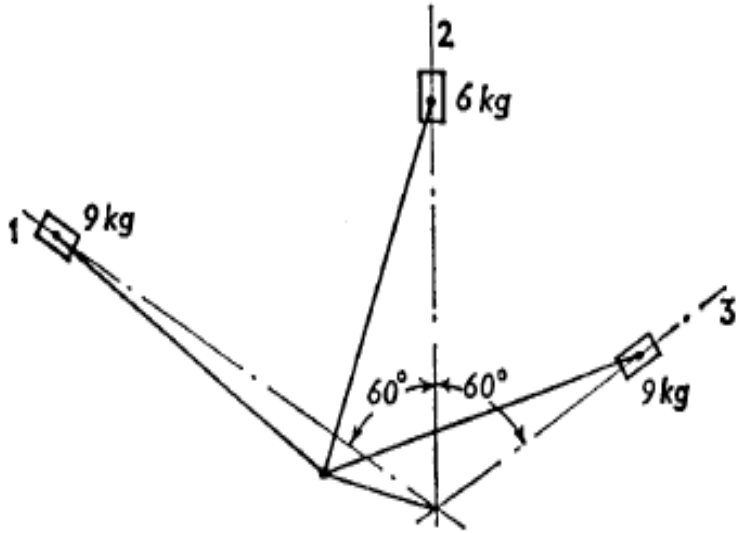
Four masses A, B, C and D are attached to a shaft and revolve in the same plane. The masses are 12 kg, 10 kg, 18 kg and 15 kg respectively and their radii of rotations are 40 mm, 50 mm, 60 mm and 30 mm. The angular position of the masses B, C and D are 60°, 135° and 270° from the mass A. Find the magnitude and position of the balancing mass at a radius of 100 mm.

# ASSIGNMENT 3

The spacing of the four cylinders A, B, C and D of a vertical in-line engine is 650 mm, 500 mm and 650 mm. The reciprocating masses of the inner cylinders B and C arc 50 kg, and their cranks are at 60° to one another; the stroke is 325 nun and the connecting rods are 600 mm long. Find the magnitudes of the reciprocating masses for the outer cylinders A and D and the relative angular positions of all the cranks if all primary forces and couples are to be balanced. What will be the maximum unbalanced secondary force acting on the base when the engine is run at 375 rev/min.

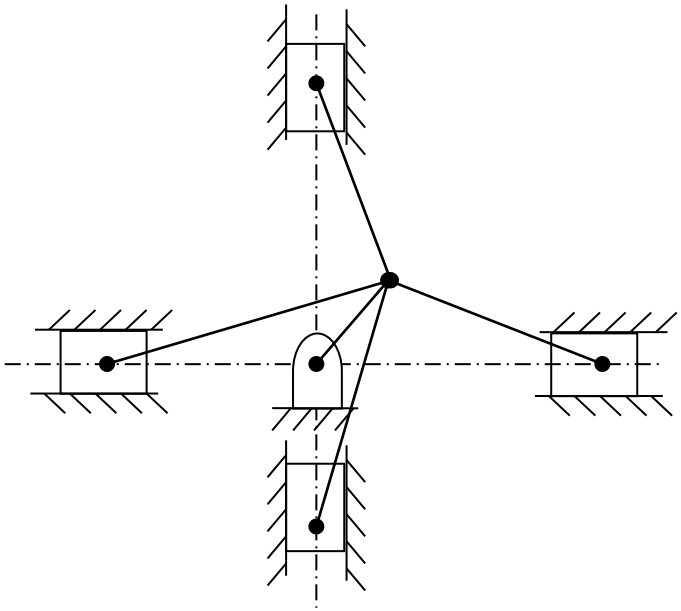
# ASSIGNMENT 4

A three-cylinder compressor is arranged as shown with the three pistons operated from a common crank. The reciprocating masses are 9 kg for cylinders 1 and 3, and 6 kg for cylinder 2. The crank radius is 40 mm; and the ratio of connecting rod to crank is 5 for each cylinder. The speed is 960 rev/min. Find the primary unbalanced force at this speed, and obtain a suitable balance mass at crank radius for vertical primary balance. Find also the unbalanced secondary force.



# ASSIGNMENT 5

The four-cylinder radial engine depicted in the Figure is an excellent engine from the point of view of dynamic balance. Show that the engine can be balanced by means of a single rotating counterweight mounted on the crankshaft, and determine the magnitude and location of such a counterweight. The crank length is *r*, the connecting rod lengths all equal *l*, and the reciprocating masses all equal *m*. The rotating masses are balanced, and all four cylinders lie in a single transverse plane.



# ASSIGNMENT 6

Investigate the state of balance of an eight-cylinder V-engine consisting of two banks of cylinders, each having four cylinders in line and both working upon one four-throw crankshaft. The centre-lines of the two banks are inclined at angles (Ø/2) on each side of the vertical plane. The relative positions of the four cranks are 0°, 90°, 180° and 270° and two connecting rods work on each crank. State the nature and amount of the total force and couple, (a) when Ø = 90°, (b) when Ø= 60° (c) Ø = 45°